Bow Valley College

Course Outline

MAT3211

Mathematics 31
Spring 2022 - Current

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MAT3211 Mathematics 31

COURSE DESCRIPTION

This is an Alberta Education credit course. Math 31 is taken as a corequisite or pre-requisite to Math 30-1. Topics include differential and integral calculus, algebraic computations of functions and graphs. This course helps prepare students who want to study university level mathematics.

REQUISITES	Complete 1 of the following • Earn a minimum grade of B in each of the following courses ◦ MAT3791 - Mathematics 30-1 (5) • Earn a minimum grade of B in each of the following courses ◦ MATH0301 - Mathematics 030-1 (5)
EQUIVALENTS	None
CREDITS	5
HOURS	100
ELIGIBLE FOR PLAR	No

COURSE LEARNING OUTCOMES

Bow Valley College is committed to ensuring our graduates can demonstrate their abilities in key areas that will make them effective citizens and encourage their development as lifelong learners. In addition to the discipline-specific skills that learners acquire in their programs, the College has identified ten learning outcomes.

College-Wide Outcomes:

1. Communication
2. Thinking Skills
3. Numeracy and Financial Literacy
4. Working with Others
5. Digital Literacy
6. Positive Attitudes and Behaviours
7. Continuous Learning
8. Health and Wellness Awareness
9. Citizenship and Intercultural Competence
10. Environmental Sustainability

#	COURSE LEARNING OUTCOME(S)	COLLEGE WIDE OUTCOMES SUPPORTED
1	Demonstrate that functions, as well as variables, can be combined, using operations, such as addition and multiplication.	1, 2, 3, 5, 7
2	Demonstrate competence in the procedures associated with the algebra of functions.	1, 2, 3, 5, 7
3	Demonstrate that functions can be transformed, and these transformations can be represented algebraically and geometrically.	1, 2, 3, 5, 7
4	Demonstrate competence in the procedures associated with the transformation of functions.	1, 2, 3, 5, 7
5	Identify that final answers may be expressed in different equivalent forms.	1, 2, 3, 5, 7
6	Demonstrate competence in the procedures associated with the construction of equivalent forms.	1, 2, 3, 5, 7
7	Identify that descriptions of change require a careful definition of limit and the precise use of limit theorems.	1, 2, 3, 5, 7
8	Demonstrate competence in the procedures associated with limits and limit theorems.	1, 2, 3, 5, 7
9	Describe the derivative of a function is a limit that can be found, using first principles.	1, 2, 3, 5, 7
10	Demonstrate competence in the procedures associated with derivatives.	1, 2, 3, 5, 7
11	Demonstrate that derivatives of more complicated functions can be found from the derivatives of simpler functions, using derivative theorems.	1, 2, 3, 5, 7
12	Demonstrate competence in the procedures associated with derivative theorems.	1, 2, 3, 5, 7
13	Explain that trigonometric functions have derivatives, and these derivatives obey the same derivative theorems as algebraic functions.	1, 2, 3, 5, 7
14	Demonstrate competence in the procedures associated with derivatives of trigonometric functions.	1, 2, 3, 5, 7
15	Describe that calculus is a powerful tool in determining maximum and minimum points and in sketching of curves.	1, 2, 3, 5, 7
16	Demonstrate competence in the procedures associated with maxima and minima.	1, 2, 3, 5, 7
17	Identify that complicated rates of change can be related to simpler rates, using the chain rule.	1, 2, 3, 5, 7
18	Demonstrate competence in the procedures associated with related rates.	1, 2, 3, 5, 7
19	Describe that the operation of finding a derivative has an inverse operation of finding an antiderivative.	1, 2, 3, 5, 7
20	Demonstrate competence in the procedures associated with antiderivatives.	1, 2, 3, 5, 7
21	Explain that the area under a curve can be expressed as the limit of a sum of smaller rectangles.	1, 2, 3, 5, 7

22	Explain that the area under a curve can be related to the antiderivative of the function defining the curve; and in turn, can use the antiderivative to determine the areas under and between curves.	1, 2, 3, 5, 7
23	Identify that velocity and acceleration are the first and second derivatives of displacement with respect to time; and that once one of the three functions is known, the other two can be found by finding derivatives or antiderivatives.	1, 2, 3, 5, 7
24	Describe that exponential and logarithmic functions have limit, derivatives and integrals that obey the same theorems as do algebraic and trigonometric functions.	1, 2, 3, 5, 7
25	Demonstrate competence in the procedures associated with the calculus of exponential and logarithmic functions.	1, 2, 3, 5, 7

COURSE MODULES AND SCHEDULE

**Course schedule subject to change, depending on delivery mode and term of study. For exact dates, please consult the Course Offering Information in Brightspace.*

WEEK/HOURS	MODULES
Weeks 1 - 4	Precalculus & Limits
Weeks 4 - 7	Derivatives & Derivative Theorems
Weeks 7 & 8	Applications of Derivatives
Weeks 8 - 12	Integrals, Integral Theorems & Integral Applications
Weeks 13- 15	Elective Component – see Course Offering Information Sheet

ASSESSMENT

COURSE LEARNING OUTCOME(S)	ASSESSMENT	WEIGHT
1 - 25	Assignments	100%
1 - 25	Tests	0%

Important: For details on each assignment and exam, please see the Course Offering Information.

PERFORMANCE STANDARDS

A minimum grade of D is required to pass this course. However, a program may require a higher grade in

this course to progress in the program or to meet specific program completion requirements.

Please consult with the program area or contact the program chair for further details. A minimum Grade Point Average of 2.0 is required for graduation.

GRADING SCHEME

Grade	Percentage	Grade Point	Description
A+	95-100	4.0	Exceptional: superior knowledge of subject matter
A	90-94	4.0	Excellent: outstanding knowledge of subject matter
A-	85-89	3.67	
B+	80-84	3.33	
B	75-79	3.0	Very Good: knowledge of subject matter generally mastered
B-	70-74	2.67	
C+	67-69	2.33	
C	64-66	2.0	Satisfactory/Acceptable: knowledge of subject matter adequately mastered
C-	60-63	1.67	
D+	57-59	1.33	
D	50-56	1.0	Minimal Pass
F	Less than 50	0.0	Fail: an unsatisfactory performance

REQUIRED LEARNING RESOURCES

Calculus: A First Course, James Stewart, et. al. ISBN-13: 978-0-07-549601-4

Additional learning resources may be found in the Course Offering Information or in Brightspace.

ADDITIONAL INFORMATION

Additional information may be found in the Course Offering Information or in Brightspace.

ACADEMIC ACCOMMODATIONS

Learners with a disability (learning, physical, and/or mental health) may qualify for academic and exam accommodations. For more information, or to apply for accommodations, learners should make an appointment with Accessibility Services in the Learner Success Services (LSS) Department. Accessibility Services can also assist learners who may be struggling with learning but do not have a formal diagnosis. To make an appointment visit LSS on the first floor of the south campus or call 403-410-1440. It is the learner's responsibility to contact Accessibility Services and request academic accommodations. For more information, please visit our website at http://www.bowvalleycollege.ca/accessibility.

INSTITUTIONAL POLICIES

Bow Valley College is committed to the highest standards of academic integrity and honesty. Learners are urged to become familiar with and uphold the following policies: Academic Honesty (500-1-7), Learner Code of Conduct, Procedures and Guidelines (500-1-1), Learner Appeals (500-1-12), Attendance (500-1-10), Grading (500-1-6), Academic Continuance and Graduation (500-1-5), and Electronic Communications (300-2-13). Audio or video recording of lectures, labs, seminars, or any other teaching and learning environment by learners is allowed only with consent of the instructor as part of an approved accommodation plan. Recorded material is to be used solely for personal study and is not being used or distributed without prior written consent from the instructor.

Turnitin:

Students may be required to submit their course work to Turnitin, a third-party service provider engaged by BVC. Turnitin identifies plagiarism by checking databases of electronic books and articles, archived webpages, and previously submitted student papers. Students acknowledge that any course work or essays submitted to Turnitin will be included as source documents in the Turnitin.com reference database, where it will be used solely to detect plagiarism. The terms that apply to a student's use of Turnitin are described on Turnitin.com.

Online Exam Proctoring:

Examinations for this course may require proctoring through an online proctoring service. Online proctoring enables online exam taking within a controlled and monitored environment, thereby enhancing academic integrity. Online proctoring may occur through a variety of methods, including but not limited to:

a. live online proctoring where a remote invigilator authenticates identity and observes completion of an exam using specialized software and recordings;
b. automated proctoring where the exam session is recorded and AI (artificial intelligence) analyzed;
c. browser lockdown that limits access to other applications, websites, copying, printing, screen capture and other functions; or
d. a combination of both live/automated proctoring and browser lockdown.

Course instructors will review recordings, analyses, and data obtained through online proctoring for academic integrity infractions. It is the student's responsibility to meet the technical, software, location, and identity verification requirements necessary to enable online proctoring.

Further details of these policies are available in the Academic Calendar and on the Bow Valley College website, bowvalleycollege.ca.

Learners are encouraged to keep a copy of this course outline for future reference.

Collection of Personal Information:

This course, including your image and voice, may be recorded and made available to you and other students taking the course section. By attending the class(es) online or in person, you consent to the collection of your personal information. If you do not wish to be recorded, please contact your instructor before starting the course/class to discuss alternative arrangements.

You may use the recordings only for educational purposes and you must not copy, share, or use the recordings for any other purpose without the instructor's express permission.

Your personal information is collected in accordance with section 33(c) of the Freedom of Information and Protection of Privacy Act (Alberta) to deliver academic programming, support learner flexibility, promote universal design for learning principles, and for purposes consistent with the course activities and outcomes. If you have any questions about the collection, disclosure, use, or protection of this information, please contact the College's Access and Privacy Officer at foip@bowvalleycollege.ca.